8/3 =2.6666666666666665

However, you can create an exact representation for rational numbers by combining together the numerator and denominator.

As we know from using functional abstractions, we can start programming productively before you have an implementation of some parts of our program. Let us begin by assuming that you already have a way of constructing a rational number from a numerator and a denominator. You also assume that, given a rational number, you have a way of selecting its numerator and its denominator component. Let us further assume that the constructor and selectors are also available.

We are using here a powerful strategy for designing programs: **'wishful thinking'**. We haven't yet said how a rational number is represented, or how the constructor and selectors should be implemented.

Note

Wishful Thinking is the formation of beliefs and making decisions according to what might be pleasing to imagine instead of by appealing to reality.

Example: An ADT for rational numbers

- - constructor

- - constructs a rational number with numerator *x*, denominator *y*

rational(*x*, *y*)

- - selector

numer(x) → returns the numerator of rational number x

denom(y) → returns the denominator of rational number y

In the above example, rational () is the constructor numer () and denom () both are selectors. In this case, selectors are declared inside the constructor but not defined.

The pseudo code for the representation of the rational number using the above constructor and selector is

```
x,y:=8,3
rational(x,y)
  numer(x)/denom(y)
- - output :  2.6666666666666665
```

2.5 Lists,Tuples

To implement the data abstraction, Programming languages like Python provides a compound structure called Pair which is made up of list or Tuple. The first way to implement pairs is with the List construct.

2.5.1 List

List is constructed by placing expressions within square brackets separated by commas. Such an expression is called a list literal. List can store multiple values. Each value can be of any type and can even be another list.

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